

Solution for Chess Tournament

Let $n = 2m + 1$. To avoid dealing with halves, we'll also multiply everything by 2: a win is 2 points, a draw is 1.

Although the problem talks about draws, everything becomes transparent once we count the other games instead. Let h_i be the number of games played by player i that were not draws, and let w_i be player i 's number of wins.

The key quantity

The total score is now twice the number of matches, i.e., $n(n - 1)$. Thus, the average score per player is $n - 1$.

Player i draws $n - 1 - h_i$ games and wins w_i , so their score is $(n - 1 - h_i) + 2w_i$. Let $t_i = 2w_i - h_i$, then t_i is the difference between this player's score and the average.

We can observe that the values t_i :

- are **distinct**, because the scores are
- **sum to zero**
- as $0 \leq w_i \leq h_i$, we know that $|t_i| \leq h_i$.

That last inequality is the key insight: a draw never moves you away from the average, only a game with a winner does. A player who finished k points away from the average must have played at least k games that had a winner.

In order to make all scores distinct, we will need to have enough games with a winner, and this will give us a tight upper bound on the number of draws possible.

The bound

Clearly, if all t_i are distinct, the smallest possible sum of their absolute values corresponds to the situation where the set of all t_i is exactly the set $E = \{-m, \dots, -1, 0, 1, \dots, m\}$.

Thus, if we look at each player and count how many of their games had a winner, the total must always be at least $\sum E = m(m + 1)$. And as each game is counted twice (once for the winner, once for the loser), **every valid tournament** must have at least $m(m + 1)/2$ games that are not draws.

Achieving the maximum number of draws

One possible construction:

- The players will end up ranked from 0 to $n - 1 = 2m$.
- The middle player m will draw all their games.
- For each i , player $m - i$ will win i of their games and player $m + i$ will lose i of their games.
- These wins and losses can easily be paired:

- Everybody in the top half wins against player $2m$.
- Everybody except player $m - 1$ also wins against player $2m - 1$.
- And so on.

What next?

We have just constructed a tournament with as many draws as possible.

Each tournament with fewer draws is also possible. Probably the easiest way to construct them is to just handwave it: start greedily turning random draws into a win + a loss such that after each update the scores are still distinct, and backtrack if you get stuck. The good states are so dense that in practice this solution quickly constructs all solutions, and it's really easy both to implement and to test — once it reaches the state with 0 draws, we know that we are done, as we must have seen all intermediate states along the way.

Below we show an alternative: one possible deterministic construction.

Move 1: rotating a triangle

Take three players where x beats y and both $x-z$ and $z-y$ are drawn. Now, rewrite the triangle as follows: x beats z , z beats y , and $x - y$ is now a draw. Clearly, the new point totals are the same as before, but now we have one fewer draw. Whenever we can apply this move, we can use it to reduce the number of draws.

Move 2: peeling off the top players

Triangle rotations alone cannot reach small values of d , because for those the players need to be farther away from the average — which means that we can no longer preserve the score totals.

One possible way how to deal with the smaller values of d is to change already the starting configuration to one with fewer draws. Instead of starting with the tournament with the most draws possible, we can choose a value b and start with a tournament such that:

- Each of the first b players has beaten everyone below them.
- Only the remaining players now have as many ties between them as possible, given that among themselves they must have different scores.

Clearly, the top b players are resolved — they will always remain top b players with mutually distinct scores, regardless of how we change the games among the rest. And thus, among the remaining players we can now freely apply the triangle rule.

It can be shown that if we start with the smallest b such that the starting tournament has d or more draws, it will always be possible to reduce their count to exactly d draws — we can always find enough triangles to rotate.